Replication Plan

Approximating Photo-z PDFs for Large Surveys

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Auto-generated Replication Blueprint

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Contents

Paper Overview

This paper introduces `qp`, a Python package for efficiently approximating, storing, and manipulating photometric redshift probability density functions (photo-z PDFs) for large astronomical surveys. The work evaluates multiple PDF approximation schemes—histograms, quantiles, samples, step functions, piecewise-linear interpolation, and Gaussian mixtures—using information-theoretic metrics (Kullback–Leibler divergence) and moments. The goal is to enable compact storage of billions of per-galaxy PDFs for surveys like LSST while preserving statistical fidelity.

Detailed Workflow

Phase 1: Environment and Package Setup

1.1 Clone the `qp` repository from GitHub.

1.2 Install dependencies: NumPy, SciPy, scikit-learn, matplotlib, pathos (for parallelism).

1.3 Verify installation by running the provided test suite.

Phase 2: Mock Photo-z PDF Generation

2.1 **Generate a library of mock photo-z PDFs.**

- Use Gaussian mixture models with varying numbers of components (1–5) to simulate realistic multi-modal photo-z PDFs.
- Sample redshift range $z \in [0, 3]$ with resolution $\Delta z = 0.01$.
- Generate ∼1000 diverse PDFs with varying widths, skewness, and multi-modality.

2.2 **Optionally use real photo-z PDFs** from public catalogs (e.g., SDSS, BPZ outputs).

Phase 3: PDF Approximation Methods

3.1 **Implement/use each approximation scheme:**

- *Histogram*: bin the PDF into N bins of uniform or variable width.
- *Quantile*: store N quantile values that summarize the CDF.
- *Samples*: draw N samples from the PDF.
- *Step function*: piecewise-constant approximation.
- *Piecewise linear*: linear interpolation between N nodes.
- *Gaussian mixture*: fit a K-component GMM to the PDF.

3.2 **Vary the compression parameter** N (number of bins/quantiles/samples/components) across a wide range.

Phase 4: Metric Computation

4.1 **Compute KL divergence** $D_{\mathrm{KL}}(p\|\hat{p})$ between original and approximated PDFs.

4.2 **Compute moment residuals**: mean, variance, skewness, kurtosis of original vs. approximated.

4.3 **Compute stacked (ensemble)** $n(z)$ **residuals**: aggregate PDFs and compare the stacked distribution.

4.4 **Record storage cost** (bytes per PDF) for each scheme at each N.

Phase 5: Results Reproduction

5.1 Reproduce metric-vs-N curves (Figures in the paper).

5.2 Reproduce storage-vs-fidelity trade-off plots.

5.3 Reproduce the comparison table of approximation schemes.

5.4 Validate by comparing to paper's stated numerical results.

Datasets

Dataset / Source	Type	Location / Notes
Mock photo-z PDFs (Gaussian mixtures)	Synthetic	Generated using `qp`'s built-in utilities or custom scripts
`qp` example/test data	Bundled test data	Included in the `qp` GitHub repository `https://github.com/aimalz/qp`
LSST DESC photo-z challenge data (optional)	Public catalog	`https://github.com/LSSTDESC/RAIL`
SDSS photometric redshift catalog (optional)	Public survey	`https://www.sdss.org/dr17/`
BPZ photo-z outputs (optional)	Public	Available via SDSS or DES data releases

Models, Tools, and Software

Programming Languages

- **Python 3.7+** — all code is in Python.

Libraries and Frameworks

Tool / Library	Version	Purpose / URL
`qp`	latest	Core package for photo-z PDF approximation `https://github.com/aimalz/qp`
NumPy	≥ 1.18	Numerical array operations
SciPy	≥ 1.5	Statistical distributions, integration, interpolation
scikit-learn	≥ 0.24	Gaussian mixture model fitting
Matplotlib	≥ 3.3	Plotting metric curves and PDF comparisons
pathos	$\geq 0.2.8$	Parallel computation for large PDF ensembles

Tool / Library	Version	Purpose / URL
Jupyter	≥ 6.0	Notebook environment for interactive analysis

Hardware Requirements

- Any modern laptop/workstation; computations are lightweight.
- For scaling tests with millions of PDFs: $\geq 16\,\text{GB}$ RAM.